Quaternionic unimodular matroids have the half-plane property

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Editorial note. Only the designated primary theorem counts. Supporting named results and other statements are uncounted. Focused reading locates and understands the published author proof; it is not a stepwise independent proof verification.

Allowed standard prerequisites (editorial).

- Quaternionic unimodular representation and quaternionic Cauchy-Binet. Formation of M(C), normalized quaternionic matrix with basis minors 0 or1 and quaternionic Cauchy-Binet are prior algebraic inputs; identifying its stable complex determinant with h_M squared remains the task.
- Stable determinant polynomials and polynomial identity principle. Determinant of a full-rank complex Gram pencil is stable, and equality on positive orthant implies polynomial equality.

1 Preliminaries

We denote by $\mathbb{N}$ and $\mathbb{N}_0$ the set of positive and nonnegative integers, respectively. In this section we let E always denote a finite set. For $i \in E$ we denote by $\delta_i \in \mathbb{R}^E$ the ith unit vector. For $x \in \mathbb{R}^E$ we write $|x| = \sum_{i \in E} |x_i|$. We further let $[n] = \{1, \dots, n\}$ and denote by $\binom{[n]}{k}$ the set of all k-element subsets of $[n]$ for every $k, n \in \mathbb{N}$. We recall the cryptomorphic definitions of M-convex sets and polymatroids.

Definition 1.1. A subset $J \subseteq \mathbb{N}_0^E$ is *M-convex* if for every $i \in E$ and every $\alpha, \beta \in J$ such that $\alpha_i > \beta_i$, there is $j \in E$ satisfying

$$\alpha_j < \beta_j \text{ and } \alpha - \delta_i + \delta_j \in J.$$

Definition 1.2. A *polymatroid* on E is a function $r \colon 2^E \to \mathbb{N}_0$ such that we have for all $S, T \subseteq E$:

(i) $r(\emptyset) = 0$,

(ii) $r(S) \leq r(T)$ if $S \subseteq T$, and

(iii) $r(S \cup T) + r(S \cap T) \leq r(S) + r(T)$.

If $J \subseteq \mathbb{N}_0^E$ is an M-convex set, then we define the function $r_J \colon 2^E \to \mathbb{N}_0$ by

$$r_J(S) = \max\{\sum_{i \in S} \alpha_i \mid \alpha \in J\}.$$

Conversely, if $r\colon 2^E \to \mathbb{N}_0$ is a polymatroid, then we define the set $J_r \subseteq \mathbb{N}_0^E$ by

$$J_r = \{x \in \mathbb{N}_0^E \mid \sum_{i\in S} x_i \leq r(S) \text{ for all } S \subseteq E \text{ and } \sum_{i\in E} x_i = r(E)\}.$$

These two constructions are inverse to each other and define bijections between the set of M-convex sets in $\mathbb{N}_0^E$ and the set of polymatroids on E, see for example [11, §4.4].

Remark 1.3. A polymatroid $r\colon 2^E \to \mathbb{N}_0$ is the rank function of a matroid M on E if and only if $r(\{i\}) \leq 1$ for all $i \in E$. In this case we have

$$J_r = \{\sum_{i\in B} \delta_i \mid B \text{ is a basis of } M\}.$$

We apply the definitions made for polymatroids to matroids by considering their rank functions as polymatroids.

1.1 Amalgamation of polymatroids

1.2 Stable polynomials

We briefly recall the different stability and real zero properties of polynomials and their relation to each other. As a general reference we recommend the survey [13].

Definition 1.4. Let $P \in \mathbb{R}[x_i \mid i \in E]$ be a polynomial.

(a) The *support* of P is the unique subset $\operatorname{supp}(P) \subseteq \mathbb{N}_0^E$ such that we can write

$$P = \sum_{\alpha\in\operatorname{supp}(P)} c_\alpha x^\alpha$$

for some non-zero $c_\alpha \in \mathbb{R}$.

(b) The polynomial P is called *stable* if for all $z \in \mathbb{C}^E$ such that $\operatorname{Im}(z_i) > 0$ for all $i \in E$ we have $P(z) \neq 0$.

(c) The polynomial P is called a *real zero polynomial* if for all $v \in \mathbb{R}^E$ the univariate polynomial $P(t \cdot v) \in \mathbb{R}[t]$ has only real zeros.

(d) Let P be homogeneous and $e \in \mathbb{R}^E$. Then P is called *hyperbolic* with respect to e if for all $v \in \mathbb{R}^E$ the univariate polynomial $P(t \cdot e + v) \in \mathbb{R}[t]$ has only real zeros.

Taking $v = 0$ in the definition of a real zero polynomial, yields that a real zero polynomial does not vanish at the origin. Similarly, a polynomial P which is hyperbolic in direction e, satisfies $P(e) \neq 0$. Stability is preserved under taking partial derivatives in coordinate directions, under setting some variables equal to each other and under scaling variables by positive scalars. We summarize the well-known connection between the above concepts.

Definition 1.5. Let $J \subseteq \mathbb{N}_0^E$ be an M-convex set.

(a) The *generating polynomial* of J is

$$h_J = \sum_{\alpha\in J} \frac{x^\alpha}{\alpha!} \in \mathbb{R}[x_i \mid i \in E]$$

where $\alpha! = \prod_{i\in E} \alpha_i!$.

(b) We say that J has the *half-plane property* if its generating polynomial is stable.

(c) We say that J has the *weak half-plane property* if there exists a homogeneous stable polynomial $P \in \mathbb{R}[x_i \mid i \in E]$ such that $J = \operatorname{supp}(P)$.

(d) We say that a polymatroid r on E has the *(weak) half-plane property* if the associated M-convex set J_r has the (weak) half-plane property.

In the remaining part, we observe that in certain nice situations, restricting the polymatroid of a homogeneous stable polynomial corresponds to plugging in zeros for some of the variables.

2 Quaternionic unimodular matroids

We recall the definition of quaternionic unimodular (QU) matroids. To this end, let $\mathbb{H}$ denote the skew field of quaternions.

Definition 2.1. Let E be a finite set and $C \subseteq \mathbb{H}^E$ a submodule of the free left $\mathbb{H}$-module $\mathbb{H}^E$. A nonzero element $x \in C$ is called an *elementary chain* of C if C does not contain a nonzero element whose support is strictly contained in the support of x. The submodule C is called *unimodular* if for every elementary chain x of C all nonzero entries of x have the same norm.

Theorem 2.2 ([14, Theorem 3.7])**.** *Let E be a finite set and $C \subseteq \mathbb{H}^E$ unimodular. The set of supports of elementary chains in C is the set of cocircuits of a matroid $M(C)$ on E.*

Definition 2.3. A matroid on a finite set E of the form $M(C)$ for some unimodular $C \subseteq \mathbb{H}^E$ is called *quaternionic unimodular (QU)*.

The goal of this section is to prove that every QU matroid has the half-plane property. This has been conjectured in [14, Conjecture 6.9].

Definition 2.4 ([14, page 219])**.** Denote by $\varphi\colon \mathbb{H} \to \mathbb{C}^{2\times 2}$ the map

$$\varphi(a + bi + cj + dk) = \begin{pmatrix} a + bi & c + di \\ -c + di & a - bi \end{pmatrix}.$$

We extend φ to matrices by applying φ entry-wise. Thus for a matrix $A \in \mathbb{H}^{n\times n}$ we obtain the $2n \times 2n$ complex matrix $\varphi(A)$. We define

$$\delta(A) \coloneqq \sqrt{|\det(\varphi(A))|}.$$

Remark 2.5. We collect some basic properties of δ.

1. For columns $a_1, \ldots, a_n \in \mathbb{H}^n$ and $\lambda \in \mathbb{R}$, we have

   $$\delta(\lambda a_1\ a_2 \ldots a_n) = |\lambda| \cdot \delta(a_1 \ldots a_n).$$

   This follows directly from the definition.

2. For matrices $A, B \in \mathbb{H}^{n\times n}$ we have $\delta(AB) = \delta(A)\delta(B)$ and $\delta(A) = \delta(A^*)$. This is [14, Lemma 5.2].

For a matrix $A \in \mathbb{H}^{m\times n}$ with $m \leq n$ and $B \subseteq [n]$ of size m we denote by $A[B]$ the $m \times m$ submatrix of A consisting of the columns indexed by B. The following is a version of the Cauchy–Binet theorem over $\mathbb{H}$.

Proposition 2.6 ([14, Theorem 5.1])**.** *Let $A \in \mathbb{H}^{m\times n}$ be a matrix over the quaternions and $m \leq n$. Then*

$$\delta(AA^*) = \sum_{\substack{B\subseteq[n]\\|B|=m}} \delta(A[B]A[B]^*).$$

Lemma 2.7. *Let E be a finite set and $C \subseteq \mathbb{H}^E$ unimodular. Let d be the rank of the matroid $M = M(C)$. There is a $d \times |E|$ matrix A over $\mathbb{H}$ whose rows form a basis of C such that for all $B \subseteq E$ of size d we have $\delta(A[B]) = 1$ if B is a basis of M and $\delta(A[B]) = 0$ otherwise.*

Proof. Let A be a matrix over $\mathbb{H}$ whose rows form a basis of C. By [14, Lemma 3.14] it has d rows, so $A \in \mathbb{H}^{d\times|E|}$. After multiplying A from the left by an invertible $d \times d$ matrix over $\mathbb{H}$, we can assume by [14, Corollary 3.26] that A is a *strong QU matrix* in the sense of [14, Definition 3.23]. Now the statement of the lemma follows from [14, Claim 5.4.1]. □

Theorem 2.8 ([14, Conjecture 6.9])**.** *Every QU matroid has the half-plane property.*

Proof. Let $E = \{1,\dots,m\}$ and $C \subseteq \mathbb{H}^m$ be unimodular. Denote by $M = M(C)$ the associated QU matroid and let $A \in \mathbb{H}^{d\times m}$ be a matrix as in Theorem 2.7. We have to show that the bases generating polynomial h_M of M is stable. This is equivalent to h_M^2 being stable. We prove this by showing that h_M^2 agrees with the stable polynomial

$$\det\left(\varphi(A)\begin{pmatrix} x_1 & & & & \\ & x_1 & & & \\ & & \ddots & & \\ & & & x_m & \\ & & & & x_m \end{pmatrix}\varphi(A)^*\right).$$

It suffices to show that these two polynomials agree on the positive orthant. We denote by $a_1,\dots,a_m \in \mathbb{H}^d$ the columns of A. Let $x \in \mathbb{R}^m_{>0}$ and write $x = (x_1^2,\dots,x_m^2)$ for $x_1,\dots,x_m \in \mathbb{R}_{>0}$. Then we have

$$h_M(x)^2 = \left(\sum_{\substack{B\subseteq E\\|B|=d}} \delta(A[B]A[B]^*)x^B\right)^2 \tag{2.1}$$

$$= \left(\sum_{\substack{B\subseteq E\\|B|=d}} \delta((x_1a_1\dots x_ma_m)[B](x_1a_1\dots x_ma_m)[B]^*)\right)^2 \tag{2.2}$$

$$= \delta((x_1a_1\dots x_ma_m)(x_1a_1\dots x_ma_m)^*)^2 \tag{2.3}$$

$$= |\det(\varphi(x_1a_1\dots x_ma_m)\varphi(x_1a_1\dots x_ma_m)^*)| \tag{2.4}$$

$$= |\det\left(\varphi(A)\begin{pmatrix} x_1^2 & & & & \\ & x_1^2 & & & \\ & & \ddots & & \\ & & & x_m^2 & \\ & & & & x_m^2 \end{pmatrix}\varphi(A)^*\right)| \tag{2.5}$$

$$= \det\left(\varphi(A)\begin{pmatrix} x_1^2 & & & & \\ & x_1^2 & & & \\ & & \ddots & & \\ & & & x_m^2 & \\ & & & & x_m^2 \end{pmatrix}\varphi(A)^*\right). \tag{2.6}$$

Here we have equality in (2) by Theorem 2.5 and Theorem 2.7. (3) holds by Theorem 2.5. For (4) we use Theorem 2.6 and (5) follows from the definition of δ. (6) and (7) are obvious. □

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